Polygenic Risk Score Portability On PDAC Data

Overview

A European-derived pancreatic cancer score, PGS000725, was applied to 2,504 individuals from five 1000 Genomes ancestry groups. All 22 variants were used after allele harmonization. A pooled top-10% cutoff selected 35.4% of AFR participants but only 0.0–2.3% in the other groups. Discrimination was modest, and an EUR-fitted model overpredicted the simulated AFR outcome rate. The PRS increased cross-validated AUC beyond age by 0.020. Genotypes and ancestry labels were real, and age and disease status were simulated. Score distributions and cutoff fractions describe the genetic inputs supplied.

Data and provenance

PGS000725 contains 22 genome-wide significant variants with beta weights and was derived from European-ancestry associations. Its PGS Catalog entry reports European discovery data from 21,536 individuals and a UK Biobank evaluation of 400,812 individuals, including 432 incident pancreatic cancer cases, with AUROC 0.639 [1–3].

The notebook was run with USE_DEMO = False. Inputs were documented as 1000 Genomes phase 3 genotypes on GRCh37, with dosage counting ALT alleles [4]. Input preparation was a public-source extraction rather than use of the Canvas-provided files. Retaining the extraction and simulation scripts is necessary for full provenance.

Proof of data from notebook Section 2

source : REAL
people : 2504
score size : 22 variants
By ancestry : AFR 661; EAS 504; EUR 503; SAS 489; AMR 347
genotypes.csv: sha256 bbd9bb17
variants.csv: sha256 2a0cf215
score.csv: sha256 a8e3a9d6
meta.csv: sha256 b582b129

Allele harmonization

All allele labels were converted to uppercase before matching. Each variant was then handled as follows:

Matched ALT: The original dosage was kept
Matched REF: The dosage was changed to 2-dosage

Matched neither: The variant was excluded

Variants missing from the genotype panel were counted separately. Before calculating the PRS, variants were placed in the same order as their weights. Sample IDs were also checked to confirm that the genotype and metadata rows matched, and no missing values were found.

Score variants	Kept as ALT	Flipped	Used total	Mismatch drops	Not found	Palindromic
22	10	12	22	0	0	3

Three A/T or C/G SNPs were flagged because allele matching alone cannot resolve strand ambiguity. No variants failed the matching rule so this does not rule out every strand or build error. Cases had a higher mean standardized PRS than controls (+0.174 versus −0.073), an internal consistency check rather than independent confirmation of alignment.

PRS distribution and the pooled cutoff

PRS was calculated as the matrix product of effect-allele dosages and published weights. The resulting score was standardized using the whole-cohort mean and standard deviation. A single cutoff was defined at the cohort 90th percentile, z = 1.427; the fraction strictly above that value was calculated within each ancestry group.

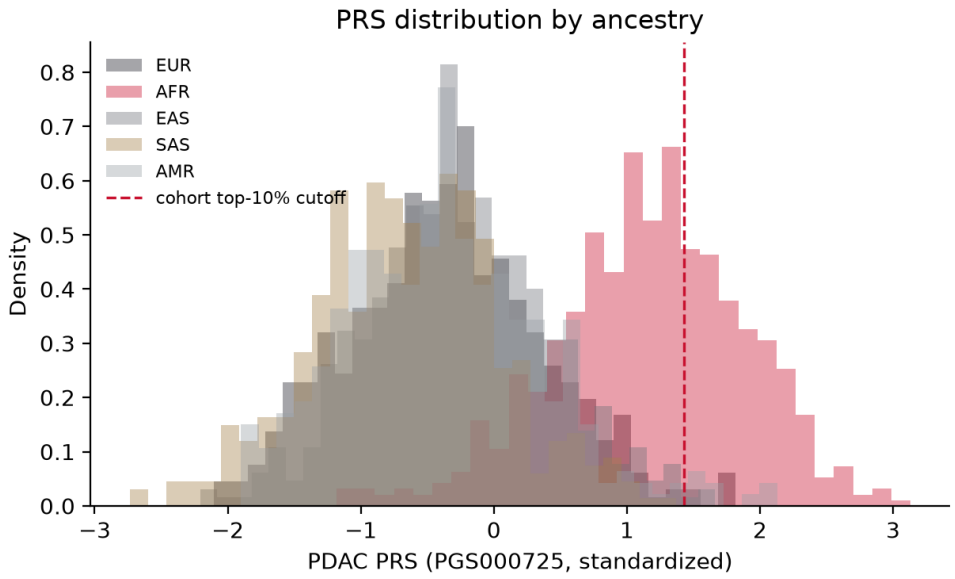


Figure 1. Standardized PRS distributions from real genotype inputs. The dashed line marks the pooled 90th-percentile cutoff. A higher score distribution does not establish a higher population disease risk.

Ancestry	n	Mean PRS z	SD	Above cutoff

EUR	503	−0.358	0.718	1.2%
AFR	661	+1.164	0.717	35.4%
EAS	504	−0.268	0.664	0.6%
SAS	489	−0.658	0.694	0.0%
AMR	347	−0.382	0.742	2.3%

Table 1. PRS summary and ancestry-specific selection rates under the same pooled threshold. SDs are within-group sample standard deviations; z values use the cohort standard deviation.

Interpretation of the shift

AFR had the highest average weighted score, while SAS had the lowest. The AFR–EUR mean difference was 1.522 cohort standard deviations. These differences arise from the observed effect-allele dosage distributions under fixed weights. They are not estimates of differences in PDAC prevalence or individual clinical risk.

Fairness of a shared threshold

The same cutoff selected very different proportions across ancestry groups. If used to decide what receives further testing, it could lead to more referrals among AFR participants while leaving nearly all SAS and EAS participants. Since average simulated outcome probabilities were similar across groups, this raises concerns about applying one pooled PRS cutoff to every.

However, selecting the top 10% overall does not mean that exactly 10% of each group should be selected. Similar average disease rates also not mean that individual risks are distributed in the same way. Fairness therefore needs to be assessed using sensitivity, false-positive rates, agreement between predicted and observed risk, and the consequences of referral decisions. Group-specific percentiles can make selection rates more equal, but they cannot improve a score's ability to distinguish cases from controls.

Discrimination and calibration

PRS-only AUC was calculated overall and within each ancestry group. Percentile 95% confidence intervals were obtained from 1,000 bootstrap resamples with seed 6900. A logistic model using PRS alone was fitted in EUR and applied to all participants. Calibration curves were formed using five within-group quantile bins of predicted probability.

Group	Cases / n	AUC with 95% CI	Mean predicted	Observed
Overall	741 / 2504	0.574 (0.548–0.597)	—	—
EUR	144 / 503	0.619 (0.566–0.672)	0.286	0.286
AFR	198 / 661	0.549 (0.503–0.597)	0.510	0.300
EAS	155 / 504	0.589 (0.533–0.638)	0.297	0.308
SAS	143 / 489	0.595 (0.546–0.650)	0.249	0.292
AMR	101 / 347	0.653 (0.593–0.710)	0.284	0.291

Table 2. Ranking performance and mean predicted versus observed simulated outcome probabilities. The pooled AUC mixes within-group and between-group comparisons.

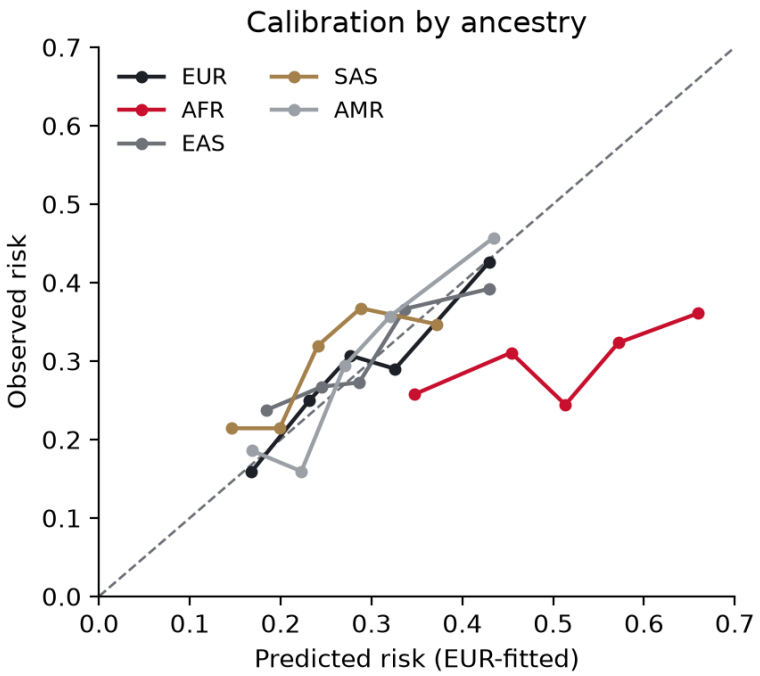


Figure 2. Calibration of the EUR-fitted model. Points below the diagonal indicate overprediction; points above it indicate underprediction. Values refer to simulated outcomes.

What the two measures reveal

AUC describes ranking, how often a case receives a higher score than a control. Calibration describes agreement between predicted probabilities and observed outcome rates. A shift shared by everyone in one group can leave within-group AUC unchanged while altering probabilities from a model transferred from another group.

AFR showed the lowest AUC (0.549), close to chance, and a mean overprediction of 0.210, or 21.0 percentage points. SAS showed a smaller mean underprediction of about 4.3 percentage points. AMR had the highest AUC point estimate, but its interval overlapped EUR's, the sampling variability is a possible explanation, and not an established cause.

EUR predictions were evaluated on the model's training sample. Agreement of its mean prediction and observed rate is expected from fitting the intercept, but does not establish calibration in new patients. Held-out or out-of-fold assessment would be needed for further invesitigation. The mean differences supplement the curves and do not summarize every aspect of calibration.

Interpretation

LD and allele frequency differences

European GWAS associations may tag causal variants through linkage disequilibrium patterns that differ across populations. Changes in LD can weaken the relationship between a scored marker and an unobserved causal variant. Allele-frequency differences also change how much variants

contribute to score means, variances and ranking. Unequal discovery sample sizes and uncertainty in estimated weights further limit transferability [5].

These are general reasons why European-derived scores can perform less well elsewhere. LD was not measured against causal variants in this analysis, so the observed AUC differences cannot be attributed directly to a measured LD mechanism. The findings also do not demonstrate that PDAC biology or causal loci are irrelevant in any ancestry group. Broad super-population labels summarize population structure and do not define uniform biological categories.

What was simulated and what was observed

According to the simulation notes, age was sampled from a normal distribution with mean 65 and SD 8 years, clipped to 40–90. Outcome probabilities were generated from a logistic model containing an age coefficient of 0.06 per year and a genetic-liability coefficient of log(1.5). Liability combined a group-centered PRS in EUR-SD units with random noise.

Assumed tagging parameters were EUR 1.00, AMR 0.79, SAS 0.79, EAS 0.71 and AFR 0.47. These should be treated as illustrative simulation assumptions motivated by portability literature, rather than validated PDAC-specific estimates. Group intercepts targeted an average outcome probability near 30%, and observed case fractions ranged from 28.6% to 30.8%. Individual probabilities were not equal.

The ancestry distribution shift and cutoff fractions were calculated from real dosages and published weights without using outcomes. In contrast, AUC, calibration and the age comparison depend on the chosen simulation. AFR calibration drift reflects both the shifted score distribution and the simulated outcome model. It is not independent evidence that real AFR patients would experience a 21-point clinical risk overestimate.

Ethics and threshold use

Justice requires attention to whether unequal discovery representation leads to unequal access to useful risk assessment. A common cutoff could produce disproportionate referrals and exclusions. Potential harms include anxiety, investigation costs, incidental findings and missed opportunities for appropriate assessment. These are plausible consequences of deployment, not harms demonstrated in this cohort.

Respect for persons requires clear communication of the score's development population, uncertainty and lack of clinical validation in this exercise. Patients should not be labelled as having high disease risk because their ancestry group has a higher raw score distribution. Privacy protections and responsible handling of genetic information are also necessary.

Accountability requires reporting performance within groups, evaluating errors and downstream decisions, and involving affected populations in validation. Potential remedies include more diverse

GWAS, multi-ancestry score development and population-appropriate recalibration. Recalibration can improve probabilities but does not by itself repair poor ranking; genetic-PC adjustment or percentile adjustment also requires validation.

Limits of this exercise

The main limitations were simulated age and outcomes, a sparse 22-variant score, modest subgroup sizes, unresolved strand ambiguity at three palindromic SNPs, and coarse ancestry categories. Bootstrap intervals quantify sampling variability conditional on these inputs, and they do not capture uncertainty in the score weights or simulation assumptions. The artificial case fraction is far above real population PDAC incidence, so the probabilities cannot be interpreted as clinical absolute risks.

Clinical covariate and incremental value

Age-only and age-plus-PRS logistic models were compared using five-fold stratified cross-validation, shuffled with seed 6900. The same fold scheme was used for both models. Comparisons were also repeated within ancestry groups.

Model	Overall	EUR	AFR	EAS	SAS	AMR
Age only	0.629	0.648	0.625	0.647	0.609	0.610
Age + PRS	0.649	0.691	0.637	0.664	0.640	0.677
AUC gain	+0.020	+0.043	+0.012	+0.018	+0.030	+0.066

Table 3. Mean cross-validated AUCs and gains calculated before rounding.

The overall gain was modest (+0.020). The largest observed subgroup gain was in AMR, followed by EUR, and the smallest was in AFR. These differences were descriptive and were not formally tested. Both age and phenotype were simulated, so this comparison does not establish incremental clinical value for real PDAC prediction. Age was standardized before splitting; a pipeline fitted within each training fold would provide stricter preprocessing separation.

Potential utility and validation needs

A validated PRS could be investigated as one component of a broader PDAC risk model or for research cohort enrichment. Such a model could combine family history, measured age, smoking, diabetes history and relevant pathogenic germline variants. The current exercise does not justify surveillance eligibility, population screening, stand-alone diagnosis or reassurance based on a low score in any ancestry group.

Real validation would require independent genotyped PDAC cases and controls across ancestries, reliable diagnoses, measured clinical covariates and sufficient cases in each group. PanScan/PanC4 or suitable biobank cohorts could support evaluation, subject to data availability and access. Representative longitudinal cohorts with follow-up and incidence data would be needed to estimate absolute risk over a defined time horizon.

Before deployment, harmonization and strand handling would need verification, followed by external assessment of discrimination, calibration, threshold-specific sensitivity, specificity and positive predictive value. Population incidence and sampling design would need to be accounted for. Prospective evaluation should show that adding the PRS improves decisions and outcomes without creating disproportionate harms. AUC improvement alone is insufficient evidence of benefit.

Conclusion

All required computational steps were completed, and the results demonstrate how a pooled score threshold can produce sharply unequal selection rates. The strongest outcome-independent findings are the PRS distribution shift and cutoff fractions. Performance metrics illustrate the assumed simulation and support careful validation rather than immediate clinical use.

AI Use Disclosure

Claude (Anthropic, 2026) was used to structure the report and to mitigate errors. All content and analysis are my own. Anthropic. (2026). Claude (Opus 4.5) [Large language model]. https://claude.ai

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